Training data

$$X_{train} = X = \begin{array}{cccc} \underline{f_1} & \underline{f_2} & \underline{f_3} & \underline{f_{14}}. \\ \end{array} \left(\begin{array}{c} 339 \times 14 \\ \\ \\ \end{array} \right) \qquad \begin{array}{c}\underline{y} \\ \left(\quad \right) \end{array} \; 339 \times 1.$$

CRIM ZON INDUS

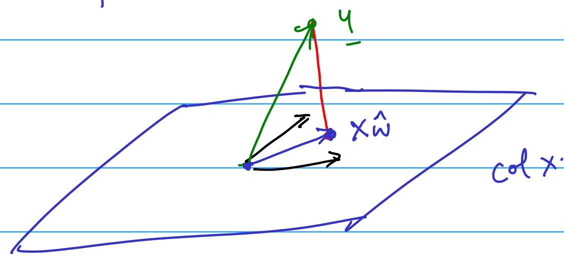
Set of all linear combinations of columns of X : $\text{Col}(X)$.

$X\underline{w}$: If $\underline{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_{14} \end{pmatrix}$ $X\underline{w} = w_1 \underline{f_1} + w_2 \underline{f_2} + \dots + w_{14} \underline{f_{14}}$

Attempt 0 Maybe $\underline{y}$ can be written as $X\underline{w}$?

Solve $\underline{y} = X\underline{w}$. $[\,X \;\; \underline{y}\,]$

Most likely there is no solution.



$\underline{y}$

$X\hat{\underline{w}}$

$y - X\hat{\underline{w}} := \underline{e}$

Red line orthogonal to plane: Red line is $\perp$ to every vector in plane.

Suppose $\mathcal{M} = \{\underline{v}_1 \dots \underline{v}_n\}$ is a maximal linearly indpt set in $\text{Col}(X)$.

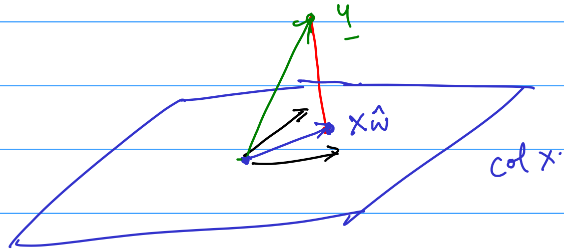
(What would be a maximal lin. ind. set in $\text{col}(X)$?)

Suppose. $\underline{e}^T \underline{v}_1 = \underline{e}^T \underline{v}_2 \dots \underline{e}^T \underline{v}_n = 0$.

Pick $\underline{x} \in \text{col}(X)$. Now $[\underline{v}_1 \dots \underline{v}_n \;\; \underline{x}]$ has the last col free.

$\therefore \underline{x} = \left[\underline{v}_1 \dots \underline{v}_n\right] \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$.

$$\underline{e}^T \underline{x} = \underline{e}^T \left[\underline{v}_1 \dots \underline{v}_n\right] \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = \left[\underline{e}^T \underline{v}_1 \dots \underline{e}^T \underline{v}_n\right] \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = 0.$$



$\underline{y}$

$X\hat{\underline{w}}$

$\underline{y} - X\hat{\underline{w}} = \underline{e}$

What is the closest point to y? It's $X\hat{w}$ such that $y - X\hat{\underline{w}}$ is orthogonal to every pivot col of X.

Assuming all cols of X are pivot cols. $y - X\hat{\underline{w}} \perp$ every col of X.

$$X = \begin{bmatrix} | & & | \\ \underline{f}_1 & \cdots & \underline{f}_{14} \\ | & & | \end{bmatrix} \qquad X^T = \begin{bmatrix} — & \underline{f}_1^T & — \\ & \vdots & \\ — & \underline{f}_{14}^T & — \end{bmatrix}$$

$$X^T \underline{e} = \begin{bmatrix} — & \underline{f}_1^T & — \\ & \vdots & \\ — & \underline{f}_{14}^T & — \end{bmatrix} \underline{e} = \begin{bmatrix} \underline{f}_1^T \underline{e} \\ \vdots \\ \underline{f}_{14}^T \underline{e} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \underline{0}.$$

$X^T \underline{e} = \underline{0}$ is the formalization that $\underline{e}$ is $\perp$ to all vectors in col(X).

$$\begin{aligned} \underline{0} = X^T \underline{e} &= X^T (\underline{y} - X\hat{\underline{w}}). \\ &= X^T \underline{y} - X^T (X \hat{\underline{w}}) \\ &= X^T \underline{y} - (X^T X) \hat{\underline{w}} \end{aligned}$$

$X : n \times p$. $\quad X^T : p \times n$

$\underset{p \times n \;\; n \times p}{X^T X} : p \times p.$ $\qquad (X^T X)^T = X^T (X^T)^T = \underline{X^T X}$

$$(AB)^T = B^T A^T$$

$$\left[\begin{pmatrix} | & | \\ \underline{a}_1 & \underline{a}_2 \\ | & | \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}\right]^T = [b_1 \underline{a}_1 + b_2 \underline{a}_2]^T = b_1 \underline{a}_1^T + b_2 \underline{a}_2^T$$

$$= (b_1 \;\; b_2) \begin{bmatrix} — & \underline{a}_1^T & — \\ — & \underline{a}_2^T & — \end{bmatrix}.$$

$$= \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}^T A^T.$$

Transpose

$$\begin{pmatrix} a & c \\ d & b \end{pmatrix} \qquad \begin{pmatrix} a & d \\ c & b \end{pmatrix}$$

Therefore

$$\underline{0} = X^T \underline{e} = X^T(\underline{y} - X\hat{\underline{w}}).$$

$$= X^T\underline{y} - X^T(X\hat{\underline{w}})$$

$$= X^T\underline{y} - (X^TX)\hat{\underline{w}}$$

$$(X^TX)\hat{\underline{w}} = X^T\underline{y}$$

vector of coefficients for our proposed linear model.

X^TX : How many pivot cols?

X : $n \times p$. X^TX : $p \times p$.

Suppose c3 of X = c1 of X + c2 of X.

$$X^TX \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = X^T\underline{0} = \underline{0}$$

c4 of X = c1 of X − c2 of X.

$$X^TX \begin{pmatrix} 1 \\ -1 \\ 0 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = X^T\underline{0} = \underline{0}.$$

Suppose $\underline{z} \in \text{null}(X)$ $\quad X\underline{z} = \underline{0}$

then $X^TX\underline{z} = X^T\underline{0} = \underline{0}$ $\quad \underline{z} \in \text{null}(X^TX)$.

$$\text{null}(X) \subseteq \text{null}(X^TX) \qquad (1)$$

Suppose $X^TX\underline{v} = \underline{0}$.

then $\underline{v}^TX^TX\underline{v} = \underline{v}^T\underline{0} = 0$

$X\underline{v}$ is a vector $\quad (X\underline{v})^T = \underline{v}^TX^T$.

$\therefore$ $(X\underline{v})^TX\underline{v} = 0$. But for any vector $\underline{u}$, $\underline{u}^T\underline{u} = \|\underline{u}\|^2$.

$\|X\underline{v}\|^2 = 0 \quad \therefore \quad X\underline{v} = 0. \quad \therefore \quad null(X^TX) \subseteq null(X) \quad — (2)$

$\therefore \quad null(X) = null(X^TX)$

In particular if X has pivot in each col, X^TX has too.

$\underset{p \times p}{(X^TX)}\, \hat{\underline{w}} = X^T\underline{y}. \qquad \hat{\underline{w}} = (X^TX)^{-1}X^T\underline{y}.$

$$\left[\begin{array}{cc} \underbrace{X^T X}_{p \text{ cols}} & X^T\underline{y} \\ p \text{ rows} & \end{array}\right].$$

To compute inverse of A. $(n \times n)$

$BA = I$ (collect all row operations to bring A to its rref)

We saw if $BA = I$ then $AB = I$.

Solve $A\underline{x}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ $\quad \left[\begin{array}{c|c} A & \begin{matrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{matrix} \end{array}\right]$ $\quad [A, \ I]$.

$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \downarrow$

$A\underline{x}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ $\quad \left[\begin{array}{c|c} A & \begin{matrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{matrix} \end{array}\right].$ $\quad [I \quad A^{-1}]$.

$\vdots$

$A\underline{x}_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$